

Spatial Equity in Healthcare: Mapping Facility Densities and Specialization Gaps Across NCR

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Target Audience

The target audience for this project consists of key stakeholders in urban development, public health, disaster management, and healthcare investment across Metro Manila:

- **Urban Planners and Spatial Analysts (MMDA & Local LGUs):** Spatial analysts and city planners rely on geographic data to identify infrastructure gaps, optimize LGU zoning, and guide future capital allocation for public services across Metro Manila.
- **Public Health Officials and Policy Makers (DOH & NCR Regional Office):** Health officials require benchmark-driven capacity evaluations to assess whether current facility ratios meet national standards and to formulate targeted healthcare access policies.
- **Emergency Response and Disaster Risk Reduction Managers (MDRRMO & OCD-NCR):** Disaster risk officers evaluate emergency readiness by identifying which high-density LGUs lack adequate high-priority specialized infrastructure, such as tertiary hospitals and birthing centers, during crises.
- **Healthcare Investors and Non-Profit Organizations:** Private developers and humanitarian groups seek data on underserved regions to strategically place new medical facilities or deploy community-based health programs where specialized care is most deficient.

Problem Statement

This study aims to evaluate the distribution, sufficiency, and type of specialization of health facilities across the 17 Local Government Units (LGUs) in the National Capital Region (NCR) relative to local population density and geographic area. Specifically, this visualization project seeks to assess healthcare emergency readiness and identify critical infrastructure disparities across Metro Manila by answering the following questions:

1. Which NCR cities fall below standard healthcare capacity benchmarks when measuring the number of health facilities per LGU?
2. How does physical health facility density compare to the population number across NCR cities, and which dense LGUs experience the highest operational deficit?

3. To what extent are LGUs in NCR dominated by general/retail health services (pharmacies, general clinics) versus high-priority specialized centers (hospitals, maternity/birthing centers, diagnostic laboratories)?

Proposed Datasets

Dataset 1: Health Facility Locations

- **Provider:** Humanitarian OpenStreetMap Team (HOT) / OpenStreetMap contributors
- **Dataset:** Philippines Health Facilities (OpenStreetMap Export)
- **Source URL:**
https://data.humdata.org/dataset/hotosm_phl_health_facilities?fbclid=IwY2xjawUsbolleHRuA2FlbQIxMQBwZG9mBHNydGMGYXBwX2lkATAAAR5n7YAkt8BlrbTQFCSvo25nlj52FoJ0Bw3z2cmhcG-qxn6_wAKEAxHwocVC0w_aem_dgtiqnmZira9mCtShIpB9Q#interactive-data
- **Usage Conditions & License:** Open Database License (ODC-ODbL) (Public)
- **Coverage:** Spatially scoped to NCR (16 cities and 1 municipality; 731 records; modified May 2026)
- **Unit:** Health facility point locations and spatial coordinates (latitude, longitude)
- **Relevant Fields:** name, amenity, healthcare, healthcare: specialty, building, operator: type, capacity: persons, addr: full, addr: city, geometry
- **Integration Keys:** addr:city (standardized municipal/city name)

Dataset 2: Population Counts

- **Provider:** Philippine Statistics Authority (PSA), 2024 Census of Population
- **Dataset:** 2024 Census of Population (POPCEN) Population Counts Declared Official by the President
- **Source URL:**
<https://psa.gov.ph/content/2024-census-population-popcen-population-counts-declared-official-president>
- **Usage Conditions & License:** Open Data with Creative Commons Attribution License (CC-BY)
- **Coverage:** 17 Local Government Units in NCR (2024 Census official headcount)
- **Unit:** Total population headcount
- **Relevant Fields:** REGION, PROVINCE, AND CITY/MUNICIPALITY, TOTAL POPULATION (pop_2024 / July 1, 2024)
- **Integration Keys:** REGION, PROVINCE, AND CITY/MUNICIPALITY

Dataset Relationship

Both datasets contain LGU administrative boundary names (addr: city in Dataset 1 and REGION, PROVINCE, AND CITY/MUNICIPALITY in Dataset 2), which serve as the primary integration keys. Initial data profiling confirmed that standardizing municipal text string variations enables direct aggregation and merging across all 17 LGUs in NCR.

Initial Python Inspection

Before starting the analysis, the group inspected both datasets in Python using pandas and numpy to understand their structure, schema, and data quality issues.

Dataset 1 (Health Facilities) contains 731 records covering health facility points across Metro Manila. Initial checking using .dtypes and .shape revealed missing spatial attributes in certain records, unformatted text entries in address fields, and missing values (NaN) in optional attributes such as capacity: persons and healthcare: specialty for smaller clinics.

Dataset 2 (2024 POPCEN Population) contains 17 LGU-level population headcounts for NCR. Inspection revealed formatting artifacts in municipal naming strings, such as thousands separators embedded in numeric text and footnote indicators attached to city labels. Additionally, administrative adjustments, specifically, the boundary shift of 10 EMBO barangays from Makati to Taguig, were identified as critical factors that required alignment during dataset integration.

The inspection verified that standardizing addr: city tags in Dataset 1 to match the cleaned CITY/MUNICIPALITY labels in Dataset 2 provides a solid join key for combining facility capacity metrics with demographic counts. This structural integration directly supports addressing Q1 (benchmarking facilities against national thresholds), Q2 (computing facility-to-population density ratios and operational deficits), and Q3 (classifying retail vs. high-priority specialized health infrastructure).

Component Mapping Table

To ensure a robust and reproducible workflow, the Python pipeline is divided into distinct, modular components. Each stage features a primary programmer responsible for implementation and a designated peer reviewer to enforce code quality and logic.

Member	Main Responsibility	Peer Reviewer
Gianela Zapata	Modular Architecture and NumPy Computations	Patricia Ann Mae Pascual
Patrick Raphael Lista	Data Acquisition and Automated Reliability	Maureen Joi Cruz
Maureen Joi Cruz	Data Cleaning, Profiling, and Quality Evidence	Patrick Raphael Lista
Patricia Ann Mae Pascual	Data Integration and Analytical Computation	Audrey Paulyne Olaybar
Audrey Paulyne Olaybar	Data Visualization and Performance Benchmarking	Gianela Zapata

Every member will contribute substantive Python code and review another member's work. Responsibilities may be adjusted as the project progresses based on operational needs and instructor feedback.

Likely Risks

- **Mismatched Keys:** Variations in city names, spacing, or boundary adjustments (e.g., EMBO barangay transfer between Makati and Taguig) that cause join errors.
Mitigation: Apply string normalization, strip text artifacts, and create an explicit LGU mapping dictionary prior to merging.
Fallback: Perform manual lookup table validation for all 17 NCR LGUs.
- **Missing Information:** High incidence of missing values (NaN) in optional OSM tags like capacity: persons and healthcare: specialty.
Mitigation: Use primary tags (amenity and healthcare) to categorize facilities into general vs. specialized centers when specific sub-tags are missing.
Fallback: Impute secondary classifications based on operator type or standard LGU-level baseline distributions.
- **Privacy:** Handling public health facility data.
Mitigation: Both datasets consist strictly of public, macro-level LGU counts and OpenStreetMap data that contain no Personally Identifiable Information (PII). Full source attribution will be maintained.
Fallback: Publish a data lineage documentation file in the repository.
- **Scope:** Risk of scope creep when expanding analysis to individual barangay levels or nationwide datasets.
Mitigation: Strictly enforce the geographic boundary to the 17 LGUs of NCR and focus solely on the three defined analytical questions.
Fallback: Shift detailed sub-category breakdowns to an appendix.

Workplan

Week	Planned Activities	Expected Output
Week 8	Finalize project scope, analytical questions, dataset metadata, initial inspection, integration keys, component mapping, and check-in proposal.	Week 8 Proposal, initial inspection notebook, sample data slices, repository setup.
Week 9	Incorporate instructor feedback, refine cleaning specifications, and implement initial text normalization for LGU join keys.	Revised proposal, cleaning script drafts, and normalized key mapping logic.

Week 10	Complete dataset cleaning for both OSM and PSA datasets. Handle missing coordinates, missing capacity attributes, and numeric formatting artifacts.	Cleaned datasets, validated data types (<code>.dtypes</code>), and data quality profiling documentation.
Week 11	Integrate Dataset 1 and Dataset 2 using standardized LGU keys. Validate merge cardinality across all 17 NCR LGUs.	Integrated dataset, automated merge scripts, and key matching validation reports.
Week 12	Conduct exploratory spatial and statistical analysis. Calculate LGU health facility densities per 10,000 population and map facility distributions.	Exploratory summary tables, preliminary density calculations, and spatial distribution outlines.
Week 13	Execute main analytical computations for Q1 (capacity benchmarks) and Q2 (population density vs. facility operational deficits).	Q1 and Q2 quantitative analysis outputs and benchmark evaluation tables.
Week 14	Execute Q3 analysis on service specialization gaps (retail vs. specialized emergency infrastructure). Run cross-variable evaluations.	Q3 analysis outputs, specialization gap matrix, and finalized analytical tables.
Week 15	Build and format the final data visualizations, spatial equity maps, and comparison charts to address all three analytical questions.	High-resolution visualization figures and visualization notebook code.
Week 16	Execute full pipeline testing, unit assertions, and reliability checks from raw data to final outputs. Address pipeline bugs.	Tested automated notebook workflow, assertion test scripts, and reliability report.
Week 17	Finalize complete documentation, repository presentation, final project deliverable package, and presentation slides.	Final project package, polished repository, and submission documentation.

Questions for the instructor: Feedback

1. Our Dataset 1 includes the capacity: person, but most of the inputs under it are NA. Should we exclude facilities with missing capacity from capacity-based analysis, or is there an appropriate way to account for the missing values?
2. If we don't have data on each facility's actual capacity, can we still assess healthcare accessibility by the number of facilities relative to population, or should we frame it as identifying potential gaps rather than determining sufficiency?

Github Repository Link: <https://github.com/giazapata/GROUP-3---FINAL-PROJECT>

Current Commit identifier: 4d5c6ecac16d91acd555a513316c5bb3c0a9d747